SARVESH V. SURVE

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PROFESSIONAL SUMMARY

Ph.D.-trained microbiologist with 8+ years of hands-on experience in microbial genetics, molecular cloning, and experimental evolution across bacterial (*Bacteroides*, *Lactiplantibacillus*, *E. coli*) and yeast systems, including recombinant protein expression, genome sequencing, and comparative genomics. Proficient in Python and R for bioinformatics; named inventor on a pending patent for a *Bacteroides*-based probiotic platform now spun out as a startup.

EDUCATION

Geisel School of Medicine at Dartmouth, Hanover, NH **August 2021-June 2027 (expected)**
Doctor of Philosophy, Microbiology and Immunology
Thesis Title: *Bacteroides* as a potential CF probiotic
Advisor: Dr. George A. O'Toole

Symbiosis International University, Pune, MH, India **August 2016-June 2018**
Master of Science, Major in Biotechnology **GPA 9.21/10.0 (Rank 1/30)**
Thesis Title: Isolation and Characterization of Lactic Acid Bacteria from Indian Foods and in vitro probiotic evaluation

University of Mumbai, Mumbai, MH, India **August 2013-June 2016**
Bachelor of Science, Major in Biotechnology **GPA 6.85/7.0 (Rank 4/30)**

RESEARCH EXPERIENCE

Geisel School of Medicine at Dartmouth, Hanover, NH **August 2021-Present**
Graduate Research Fellow, PI: Dr. George A. O'Toole, Geisel School of Medicine

- Evolved probiotic strains of *Bacteroides* to have better fitness in cystic fibrosis (CF)-like environments
- Discovered a new species of *Streptococcus* and managed a >3000 CF sample repository along with a collection of ~1200 clinical isolates of various gut species
- Responsible for project management, vendor negotiations, experiments and data analysis, and writing reports and manuscripts
- Published two first-author research articles (7 publications), with additional manuscripts in preparation
- Contributing to a patent application based on thesis research (WO2026/169644)

Marine Biological Laboratory, Woods Hole, MA **July 2025-August 2025**
Student Researcher – Microbial Diversity Course

- Conducted an independent project titled "Spatial and Genomic Ecology of *Gracilibacteria* in Green Sulfur Bacteria Enrichment Cultures Using High-Resolution Imaging and Metagenomics"
- First-ever culturing of Gracilibacteria, part of microbial dark matter
- Trained in advanced microbial ecology, microscopy, and sequencing methods

Symbiosis School of Biological Sciences, Pune, MH, India **October 2018-June 2021**
Senior Research Fellow, PI: Dr. Ram Kulkarni

- Isolated two unique *L. plantarum* strains and characterized their probiotic potential. Used these strains to evolve tolerance to inhibitory chemicals for use in lignocellulosic fermentations
- Molecular cloning and characterization of novel polyketide synthases, O-methyltransferases, hydroxylases, and sesquiterpene oxygenases from *Aquilaria malaccensis* in *E. coli* and yeast systems
- Responsible for performing and troubleshooting experiments, analyzing data, writing reports and manuscripts, and training interns
- Published three first-author papers (4 publications). Generated data and helped conceptualize the MJRP grant that was awarded to Dr. Ram Kulkarni

Institute of Nuclear Medicine and Allied Sciences, Delhi, India **May 2017-July 2017**
Summer Research Fellow, PI: Dr. Paban Kumar Agrawala

- Internship awarded as part of accepting the Summer Research Fellowship
- Studied DNA break generation in sperm cells of irradiated C57BL/6 mice

AWARDS AND HONORS

Outstanding Presentation ***September 2026***
2026 Microbial Dark Matter Symposium

Awarded for the best presentation at the symposium hosted by Zymo Research

M2P2 Scholarship ***August 2025***

Department of Microbiology and Immunology, Dartmouth College

Awarded for attending off-campus courses during Ph.D. Covers the cost of tuition, room, board, or travel.

DST INSPIRE Fellowship **December 2018**

Ministry of Science & Technology, Government of India

Awarded to the Rank 1 student at each university in India to join a funded Ph.D. program (*did not accept*)

PG Merit Scholarship **December 2016-June 2018**

Symbiosis International University

Awarded to the Rank 1 student in the M.Sc. program. The award includes a 50% tuition waiver for all the semesters of the master's degree

CSIR-UGC NET-LS **June 2018**

Ministry of Science & Technology, Government of India

Highly competitive exam in STEM to qualify for research fellowship positions in India. *All India Rank 22*

Summer Research Fellowship **May 2017-July 2017**

Indian Academy of Sciences

National fellowship awarded by the Indian Academy of Sciences to early-career students for experiencing STEM research in prestigious institutes across India

Veronica Rodriguez Award **December 2015**

Tata Institute of Fundamental Research

Awarded to members of Collaborative Undergraduate Biology Education (CUBE) for leadership and excellence

PUBLICATIONS

- **Surve, S. V.,** Sanville, J. L., Madan, J. C., & O'Toole, G. A. (2026). Longitudinal gut metagenomic remodeling in persons with cystic fibrosis before, during, and after CFTR modulator therapy. *In prep*.
- **Surve, S. V.**, & O'Toole, G. A. (2026). The CF Kids Cohort: A 17-Year Longitudinal Resource for Gut Microbiome Research. *In prep* (Review)
- **Surve, S. V.**, Eggins, S., Day, L. A., Huang, Y., Kahn, A., Charles-Orszag, A., Yang, J. A., O'Toole, G. A., Dawson, S. C., & Spear, J. R. (2026). Spatial and genomic ecology of *Gracilibacteria* in green sulfur bacteria enrichment. *In prep.*
- Abrahamson, E. A., **Surve, S. V.**, & O'Toole, G. A. (2026). *Streptococcus ruoffii* sp. nov., isolated from the oesophagus of an adult male heterozygous for the CF allele. *In review.*
- Sharma, S., **Surve, S. V.**, Fernandes, J. M., Mahale, V., Vijayvargiya, M., & Kulkarni, R. (2026). Adaptive laboratory evolution of *Lactiplantibacillus plantarum* enhances furan tolerance and lactic acid production in lignocellulosic hydrolysate. *Bioresources and Bioprocessing* Link
- **Surve, S. V.**, Valls, R. A., Barrack, K. E., Gwilt, L. L., Gardner, T. B., & O'Toole, G. A. (2026). The regional landscape of the human colon culturome in health and cystic fibrosis. *Microbiology Spectrum* Link
- **Surve, S. V.**, Valls, R. A., & O'Toole, G. A. (2026). Genome sequences of 109 gut bacteria isolated from colonoscopy samples of individuals with and without cystic fibrosis. *Microbiology Resource Announcements* Link
- Barrack, K. E., **Surve, S. V.**, Bezerra, A. V., Murphy, C. E., Soucy, S. M., Valls, R. A., & O'Toole, G. A. (2026). A genotoxin associated with colorectal cancer linked to gut dysbiosis in children with cystic fibrosis. *Journal of Bacteriology* Link
- Valls, R. A., Barrack, K. E., **Surve, S. V.**, Bliska, J. B., & O'Toole, G. A. (2026). Sex and intestinal-region-specific *Bacteroides* regulation of host response to pulmonary challenge. *mBio* Link
- Carmichael, M. M., Valls, R. A., Soucy, S., Sanville, J. L., Madan, J. C., **Surve, S. V.**, Sundrud, M. S., & O'Toole, G. A. (2025). Profiling bile acids in the stools of humans and animal models of cystic fibrosis. *Microbiology Spectrum* Link
- **Surve, S. V.**, Rajkhowa, R., Vyas, P., Vijayvargiya, M., Khare, S., & Kulkarni, R. (2024). Transcriptomic analysis identifies a furfural reductase in *Lactiplantibacillus plantarum*. *Biocatalysis and Agricultural Biotechnology* Link
- **Surve, S. V.**, Shinde, D. B., Fernandes, J. M., Sharma, S., Vijayvargiya, M., Kadam, K., & Kulkarni, R. (2024). Laboratory domestication of *Lactiplantibacillus plantarum* alters some phenotypic traits but causes non-novel genomic impact. *Journal of Applied Microbiology* Link
- Barrack, K. E., Hampton, T. H., Valls, R. A., **Surve, S. V.**, Gardner, T. B., Sanville, J. L., Madan, J. C., & O'Toole, G. A. (2024). An *in vitro* medium for modeling gut dysbiosis associated with cystic fibrosis. *Journal of Bacteriology* Link
- **Surve, S.**, Shinde, D. B., & Kulkarni, R. (2022). Isolation, characterization, and comparative genomics of potentially probiotic *Lactiplantibacillus plantarum* strains from Indian foods. *Scientific Reports* Link
  (**Collection**: Top 100 in Microbiology – 2022)
- Haritwal, T., Gupta, N., Tiwari, M., **Surve, S.**, & Agrawala, P. K. (2017). Radiation countermeasures: current status. *Defense Journal of Life Science* (Review)

CONFERENCES/PRESENTATIONS/POSTERS

- ***Microbiome 2026*** **October 27-31, 2026**
  Cold Spring Harbor Laboratory, New York
  Will be presenting a poster titled "Longitudinal gut metagenomic remodeling in persons with cystic fibrosis before, during, and after CFTR modulator therapy"
- ***Microbial Dark Matter Symposium 2026*** **September 16-17, 2026**
  Laguna Beach, California
  Presented a poster titled "Episymbiosis of Gracilibacteria with Green Sulfur Bacteria in a tidal marsh enrichment" – Winner of Outstanding Presentation award
- ***2026 Molecular Genetics of Bacteria and Phages Meeting*** **August 3-7, 2026**
  Madison, Wisconsin
  Presented a poster titled "Multilayered genomic adaptation enables *Bacteroides* survival in the cystic fibrosis gut environment" – Top 12 out of 330 posters presented at the meeting
- ***69th Wind River Conference on Prokaryotic Biology*** **June 2-6, 2025**
  Breckenridge, Colorado
  Presented a seminar titled "Mechanisms of Bile Acid Resistance in *Bacteroides*: A Role of DGRs"
- ***Dartmouth Cystic Fibrosis Research Center (DartCF)* Annual Retreat** **April 28-29, 2025**
  Hanover, New Hampshire
  Presented a seminar in a session titled "Multi-organ crosstalk"
- ***Microbiology and Molecular Pathogenesis Program (M2P2) Retreat*** **February 13-14, 2025**
  Fairlee, Vermont
  Presented a seminar titled "*Bacteroides* as a Potential CF Probiotic"
- ***Dartmouth Cystic Fibrosis Research Center (DartCF) Seminar Series*** **January 15, 2025**
  Hanover, New Hampshire
  Presented a seminar titled "*Bacteroides* as a Potential CF Probiotic"
- ***BIO International Convention*** **June 3-5, 2024**
  San Diego, California
- ***North American Cystic Fibrosis Conference*** **November 2-4, 2023**
  Phoenix, Arizona
  Presented a poster titled "*Bacteroides* as a potential cystic fibrosis probiotic"
  - **Surve, S.**, Valls, R., Barrack, K., Gardner, T., & O'Toole, G. (2023). 521 Bacteroides as a potential cystic fibrosis probiotic. *Journal of Cystic Fibrosis*, *22*, S275–S276. (Abstract)
  - Barrack, K., Valls, R., **Surve, S.**, Hampton, T., & O'Toole, G. (2023). 520 Developing an in vitro model of the cystic fibrosis gut. *Journal of Cystic Fibrosis*, *22*, S275. (Abstract)
- ***ASM Conference on Biofilms*** **November 13-17, 2022**
  Charlotte, North Carolina
- ***North American Cystic Fibrosis Conference*** **November 3-5, 2022**
  Philadelphia, Pennsylvania
- ***BESCON 2019*** **October 18-19, 2019**
  Chennai, India
  Presented a poster titled "Adaptive evolution of *Lactobacillus plantarum* to develop tolerance towards furfural and 5-hydroxymethylfurfural"
- ***VES College Biotechnology Seminar Series*** **September 2015**
  Mumbai, India
  Presented a seminar titled "*Human Genome Project and its Impact on Biology*"

LEADERSHIP AND ACTIVITIES

Graduate Student Council, Dartmouth College, Hanover **August 2022-May 2023**
Student Life Chair

- Member of the Executive Board of the Graduate Student Council
- Advocated for better public transport around Dartmouth
- Organized a monthly event series titled "Mental Health Mondays"

BioRender, Toronto, Canada (Remote) **August 2022-May 2023**
Campus Ambassador – Dartmouth College

- Initiating and maintaining the Dartmouth College and BioRender relations
- Strategizing the campus-wide licensing of the services
- Providing illustration resources for the Dartmouth community

TEACHING AND MENTORSHIP

Teaching

- ***Teaching Assistant****, BIO46 Microbiology, Dartmouth College – Spring 2023*
  Assisted in course instruction, grading, and facilitated small-group learning sessions
- ***Co-lead****, Journal Club (MICR265) – Geisel School of Medicine – Winter 2025*
  Co-organized and led discussions on microbial adaptations to complex environments

Mentorship

- ***Undergraduate Researchers (Dartmouth College)***
  - *E. Axel Abrahamson (May 2024–Present)*
  - *Citlalti V. Gaeta (Feb 2023–May 2024)*
- ***Rotation Students (MCB Ph.D. Program, Dartmouth College)***
  - *Kayla Ramirez Soto (Mar 2024–May 2024)*
  - *Natalie Culler (Dec 2025–Mar 2026)*
- ***M.Sc. Thesis Students, Symbiosis International University***
  - *Sahara Sarang (Dec 2020–May 2021)*
  - *Joyleen Fernandes (Dec 2019–May 2020)*
  - *Shruti Gaikwad (Dec 2020–May 2021)*
  - *Armaan Mistry (Dec 2019–May 2020)*

SKILLS AND INTERESTS

- Computer languages: Proficient in R, Python, Linux command line; working knowledge of bash scripting
- Bioinformatics: Genome assembly and annotation, variation analysis, comparative genomics, phylogeny, transcriptomics, microbiome analysis
- Languages: Marathi (fluent), Hindi (conversational), Sanskrit (basic)
- Interests: Formula 1, data analytics and visualization, science history, and currently visiting all the US national parks

ADDITIONAL EXPERIENCE

- ***NSF's Interior Northeast I-Corps*** **May 2024-June 2024**
  Cornell University, New York (Remote)
  Performed customer discovery for a potential startup based on my graduate research
- ***Graduate Rotations, MCB Ph.D. Program – Dartmouth College (2021–2022)***
  - ***Effect of indole-producing Bacteroides in CF gut signaling*** **March 2022-May 2022**
    Rotation 3, MCB Ph.D. Program; Advisor: Dr. Benjamin D. Ross
  - ***CF vs. non-CF: Microbiome analysis of colonoscopy samples*** **December 2021-March 2022**
    Rotation 2, MCB Ph.D. Program; Advisor: Dr. George A. O'Toole
  - ***Effect of Histone H3-tail on the nuclear cycles in Drosophila embryogenesis*** **August 2021-November 2021**
    Rotation 1, MCB Ph.D. Program; Advisor: Dr. Amanda A. Amodeo
- ***Medical Lab Technician*** **April 2015-May 2015**
  H.J. Doshi Hospital, Mumbai, India
  Clinical microbiology lab shadow